

Junjie Fei

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🌐 feielysia.github.io  [Google Scholar](#)  [LinkedIn](#)

Research Interest

- ◇ **Long Visual Context Modeling:** Long Video [1], Large-Scale Documents [2]
- ◇ **Vision-Language:** Zero-Shot / Interactive Captioning, MLLMs [3], [4], [5]

Education

- **King Abdullah University of Science and Technology**, Saudi Arabia. Aug. 2024 – present
Ph.D. in Computer Science, GPA: 3.89/4.0
Supervised by Prof. Mohamed Elhoseiny.
- **Xiamen University**, China. Sept. 2020 – Jun. 2023
M.E. in Electromagnetic Field and Microwave Technology, GPA: 3.59 / 4.00
Thesis title: Fast Electromagnetic Imaging Based on Deep Learning
- **Chongqing University**, China. Sept. 2016 – Jun. 2020
B.E. in Communication Engineering, GPA: 3.17 / 4.00

Experience

- ◇  **Research Scientist Intern**, Meta. Sept. 2025 - Feb. 2026, United States
Dr. Chencheng Zhu, XR Core AI, Burlingame
Small VLM-based smart compressors for long video understanding
- ◇  **Visiting Scholar**, KAUST. Jan. 2024 - May 2024, Saudi Arabia
Vision CAIR Research Group hosted by Prof. Mohamed Elhoseiny
3D Multimodal LLM for part-aware grounded description
- ◇  **Visiting Scholar / Research Assistant**, SUSTech. Oct. 2022 - Jan. 2024, China
Visual Intelligence & Perception (VIP) Lab hosted by Prof. Feng Zheng
Zero-Shot Image Captioning & Caption Anything

Publications

6 first/co-first authored papers (3 top-tier publications), and 4 collaborations.

- 1 **Junjie Fei** et al., “Small vision-language models are smart compressors for long video understanding,” *arXiv*, 2026, *Work done at Meta*.
- 2 J. Chen*, D. Xu*, **Junjie Fei***, C.-M. Feng, and M. Elhoseiny, “Document haystacks: Vision-language reasoning over piles of 1000+ documents,” [CVPR 2025].
- 3 T. Wang*, J. Zhang*, **Junjie Fei***, H. Zheng, Y. Tang, Z. Li, M. Gao, and S. Zhao, “Caption anything: Interactive image description with diverse multimodal controls,” *arXiv*, 2023, *GitHub 1.8k+ stars*.
- 4 **Junjie Fei***, T. Wang*, J. Zhang, Z. He, C. Wang, and F. Zheng, “Transferable decoding with visual entities for zero-shot image captioning,” [ICCV 2023].
- 5 M. Ahmed*, **Junjie Fei***, J. Ding, E. M. Bakr, and M. Elhoseiny, “Kestrel: 3d multimodal llm for part-aware grounded description,” [ICCV 2025].
- 6 M. Zhuge et al., “Neural computers,” *arXiv*, 2026.
- 7 S. Chowdhury, M. Elmoghany, Y. Abeyasinghe, **Junjie Fei**, S. Nag, S. Khan, M. Elhoseiny, and D. Manocha, “Magnet: A multi-agent framework for finding audio-visual needles by reasoning over multi-video haystacks,” [NeurIPS 2025].

- 8 Z. Yang, J. Chen, D. Xu, **Junjie Fei**, X. Shen, L. Zhao, C.-M. Feng, and M. Elhoseiny, "Wikiautogen: Towards multi-modal wikipedia-style article generation," **[ICCV 2025]**.
- 9 F. Han, M. Zhong, and **Junjie Fei**, "Hybrid microwave imaging of 3-d objects using lsm and bim aided by a cnn u-net," 2022, **[IEEE TGRS]**.
- 10 **Junjie Fei**, Y. Chen, M. Zhong, and F. Han, "Fast 3-d electromagnetic full-wave inversion of dielectric anisotropic objects based on resu-net enhanced by variational born iterative method," 2022, **[IEEE TAP]**.

Academic Services

- ◇ **Conference reviewer**, CVPR, ECCV, ICLR, ICML, NeurIPS
- ◇ **Journal reviewer**, IEEE TMM, Neurocomputing, CVIU
- ◇ **Teaching Assistant**, Xiamen University Electronic Circuit

Skills

- ◇ **Languages**: Chinese, English (Fluent).
- ◇ **Coding**: Python, C/C++, $\text{\LaTeX}$.
- ◇ **Software**: Photoshop, Lightroom.

Awards

- ◇ KAUST Graduate Scholarship. 2024 - present
- ◇ Cai Wenzhong Scholarship (Second Class), Xiamen University. 2023
- ◇ Graduate Academic Scholarship, Xiamen University. 2020 - 2023
- ◇ Successful Completion, The 10th Student Research Training Program (SRTP), Chongqing University. 2019
- ◇ Second Prize, The 10th Chinese Mathematics Competition (Non-Mathematics Majors). 2018